

WAHOO, NE, USA

WMO: 720942

Lat: 41.241N

Lon: 96.594W

Elev: 1224

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 11-19

WBAN: 00323

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) 2.7	(d) -10.9	(e) 3.2	(f) 0.6	(g) -6.3	(h) 4.1	(i) 4.0	(j) 35.1	(k) 28.2	(l) 31.4	(m) 29.6	(n) 7.0	(o) 330	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 95.3	(d) 73.0	(e) 91.9	(f) 73.2	(g) 89.5	(h) 73.1	(i) 78.7	(j) 88.2	(k) 77.1	(l) 86.6	(m) 75.4	(n) 84.8	(o) 12.3	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	143.2	84.5	74.1	133.3	82.4	72.6	126.6	81.0	43.2	88.2	41.3	86.8	39.7	84.9	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 27.3	(b) 24.4	(c) 20.9	(e) -10.8	(f) 99.4	(g) 4.1	(h) 2.9	(i) -13.8	(j) 101.5	(k) -16.2	(l) 103.2	(m) -18.5	(n) 104.9	(o) -21.5	(p) 107.0		
(4) 27.3	(5) 24.4	(6) 20.9	(7) -11.3	(8) 82.2	(9) 4.1	(10) 1.6	(11) -14.2	(12) 83.3	(13) -16.6	(14) 84.2	(15) -18.9	(16) 85.1	(17) -21.9	(18) 86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	25.5	26.6	41.4	51.3	63.5	74.0	76.5	72.6	67.3	53.3	39.3	29.1		
	DBStd	20.62	11.42	12.33	12.69	9.90	9.23	6.21	5.81	5.09	8.23	8.93	10.34	10.92		
	HDD50	2910	758	656	326	103	10	0	0	0	2	68	339	648		
	HDD65	5934	1223	1074	737	418	142	6	2	4	71	373	772	1112		
	CDD50	3588	0	2	58	143	429	721	823	702	521	171	17	1		
	CDD65	1136	0	0	4	8	95	277	360	241	140	11	0	0		
	CDH74	11493	0	0	80	167	1062	2864	3733	2017	1395	168	7	0		
	CDH80	4658	0	0	19	47	412	1216	1664	737	532	31	0	0		
(14) Wind	WSAvg	9.2	9.8	10.2	11.0	12.0	10.4	9.5	6.4	6.0	7.4	9.1	9.9	9.4		
(15) Precipitation	PrecAvg	30.2	0.7	0.9	1.6	3.0	4.7	4.9	3.5	3.9	2.7	2.0	1.4	0.9		
	PrecMax	40.4	1.7	2.5	3.5	5.1	8.6	10.8	10.6	11.5	7.1	5.6	3.0	2.4		
	PrecMin	20.2	0.2	0.1	0.0	0.5	1.3	1.3	0.7	0.5	0.6	0.1	0.0	0.0		
	PrecStd	5.3	0.4	0.6	1.0	1.0	2.0	2.3	2.2	2.6	1.6	1.4	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	69.0	81.8	86.2	96.1	97.5	99.4	97.1	94.5	83.7	73.7	60.0		
		MCWB	45.2	51.2	60.8	61.8	65.0	72.7	74.4	72.6	71.1	62.3	56.4	47.2		
	2%	DB	51.0	56.6	74.9	78.3	89.5	93.7	95.2	91.3	90.2	78.8	65.8	54.0		
		MCWB	41.7	46.9	55.0	58.1	67.3	72.5	75.9	74.0	72.2	61.0	53.3	46.5		
	5%	DB	45.6	50.3	68.1	73.3	84.3	90.3	91.8	87.1	86.4	74.3	60.5	48.8		
		MCWB	38.0	42.6	53.5	56.1	64.4	71.7	76.1	74.3	71.5	58.4	49.2	41.7		
	10%	DB	41.0	43.2	61.9	69.0	80.1	87.1	88.9	83.9	82.1	69.4	55.5	43.7		
		MCWB	36.0	38.2	51.3	54.5	63.5	70.5	75.7	72.8	68.5	56.8	46.7	38.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	53.7	63.6	64.6	72.5	79.0	81.4	80.0	77.0	68.1	60.1	52.6		
		MCDB	57.1	60.3	76.6	77.4	83.7	90.0	91.0	88.5	86.9	75.8	67.8	55.5		
	2%	WB	41.9	46.9	59.7	61.2	69.4	76.7	79.6	77.4	75.0	64.4	54.5	46.0		
		MCDB	49.9	56.7	68.9	73.5	80.5	87.0	89.3	85.0	84.3	73.3	64.2	52.6		
	5%	WB	39.1	42.4	55.5	59.2	67.4	74.9	77.9	75.5	73.4	61.5	50.6	42.1		
		MCDB	44.9	49.6	67.0	69.5	80.3	85.5	87.8	83.8	82.4	69.3	59.2	48.0		
	10%	WB	36.5	38.5	51.2	56.8	65.7	73.1	76.3	73.7	71.8	58.5	46.9	38.7		
		MCDB	40.6	43.4	64.0	66.5	78.1	83.7	86.0	81.5	79.3	66.5	56.0	43.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	19.0	23.0	23.3	21.8	20.8	20.6	20.1	23.8	23.6	22.4	17.7		
		MCDBR	27.0	28.6	35.4	32.3	28.5	24.3	23.7	25.3	27.2	31.0	30.9	26.4		
		MCWBR	18.6	18.5	18.7	15.8	10.0	9.1	10.1	11.4	11.1	15.5	18.1	18.2		
		MCDBR	24.7	27.3	28.9	25.8	23.5	21.1	20.7	21.2	22.4	24.7	28.2	24.3		
	5% WB	MCWBR	17.3	18.3	16.6	14.3	10.2	10.8	10.5	11.2	11.0	14.6	18.0	17.6		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.309	0.330	0.386	0.409	0.418	0.425	0.417	0.378	0.331	0.295	0.282		
	taud		2.455	2.402	2.394	2.251	2.261	2.296	2.276	2.298	2.361	2.483	2.505	2.462		
	Ebn at Noon		280	286	292	280	276	272	269	268	272	275	272	269		
	Edh at Noon		24	30	34	42	43	41	42	40	35	27	23	22		
(44) All-Sky Solar Radiation	RadAvg		648	941	1223	1570	1848	2067	2074	1789	1468	1037	728	545		
	RadStd		59	56	107	108	137	149	128	115	94	117	49	54		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page